Practical #1

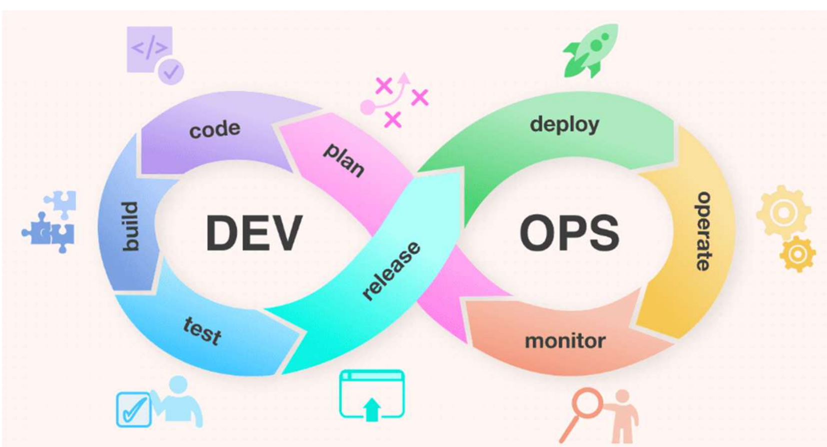
Practical Title:
To understand the DevOps culture, lifecycle, and benefits in modern software development

Theory:

1. What is DevOps?
DevOps is a combination of **Development (Dev)** and **Operations (Ops)** that aims to improve collaboration between teams to deliver software faster and more reliably. It focuses on **automation, continuous integration, and continuous delivery (CI/CD)**.

2. Key Principles of DevOps"
Collaboration – Developers and operations teams work together
Automation – Automate testing, building, and deployment
Continuous Integration (CI) – Frequent code integration
Continuous Delivery (CD) – Faster and reliable releases
Monitoring & Feedback – Continuous system tracking and improvement

3. DevOps Lifecycle



Phases Explanation

1. Planning:
Define project goals, requirements, and tasks.
2. Development:
Developers write and manage code using tools like Git.
3. Building:
Compile code and create executable files or packages.

4. Testing:
Perform automated/manual testing to find bugs.
5. Release:
Prepare software for deployment.
6. Deployment:
Deploy application to production environment.
7. Monitoring:
Track performance and fix issues continuously.

Release:
Deployment:
Monitoring:

Activity / Task:

Task 1:
Search and write one real-world company using DevOps
Company Name: Netflix

How DevOps is used:

- Uses **CI/CD pipelines** for fast updates
- Deploys code frequently (thousands of times daily)
- Uses automation tools and cloud (AWS)
- Continuous monitoring for performance and failures

Task 2:
Difference between Traditional Development vs DevOps

Feature	**Traditional Development**	**DevOps**
Team Structure	Separate teams	Collaborative teams
Deployment	Slow	Fast
Testing	Late stage	Continuous
Automation	Less	High
Feedback	Delayed	Continuous

Advantages of DevOps

1. Faster software delivery
2. Improved collaboration
3. Reduced errors and bugs

4. Continuous feedback and improvement
5. Better customer satisfaction

Challenges in DevOps

1. Requires cultural change in organization
2. Tool integration complexity
3. Skilled professionals needed